

Inhomogeneous-media benchmarks for `lebedev_em`

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Abstract

We document two inhomogeneous-media benchmarks for the `lebedev_em` solver, an implementation of the Lebedev staggered-grid finite-difference scheme of Davydycheva, Druskin & Habashy [1] (DDH03). The first is a two-half-space model with a $10\times$ conductivity contrast, validated against the exact Sommerfeld-integral solution: the four-cluster Lebedev average attains $\approx 1\%$ accuracy in the conductive half-space, and the cluster averaging reduces the single-cluster error by a factor of five. The second reproduces the self-consistency convergence test of DDH03 Figure 7 — a borehole with invasion crossing a 60° dipping anisotropic layer — where solutions on optimal geometric grids with $k = 3, \dots, 6$ transverse steps are graphically indistinguishable, and the $k = 5$ solution agrees with the $k = 6$ reference to within 1.5% at every receiver. Both benchmarks were run on the repository state that includes the July 2026 correctness fixes.

1 Background

The solver discretizes the frequency-domain Maxwell system (time convention $e^{-i\omega t}$)

$$\nabla \times \mathbf{E} = i\omega\mu\mathbf{H}, \quad \nabla \times \mathbf{H} = \dot{\sigma} \mathbf{E} + \mathbf{J}, \quad \dot{\sigma} = \sigma - i\omega\varepsilon, \quad (1)$$

on Lebedev’s staggered grid, which splits into four Yee-like clusters carrying mixed electric/magnetic boundary conditions; averaging the four cluster solutions cancels both the source-discretization and domain-truncation errors (DDH03, §“Superconvergence”). Heterogeneous and anisotropic media are handled by sub-cell homogenization (sequential nodal homogenization, after Moskow et al. [2]), and the transverse directions use the optimal geometric grids of Ingerman–Druskin–Knizhnerman with progression factor $\alpha = \exp(\gamma\pi/\sqrt{k})$, $\gamma = 1/\sqrt{2}$.

Both benchmarks below exercise the full inhomogeneous pipeline: media builders, homogenization at material interfaces, the four-cluster solve with per-cluster mixed boundary conditions, and the interpolate-then-average field extraction of DDH03 eq. (7).

2 Benchmark 1: two half-spaces, $10\times$ contrast

Model. A unit vertical electric dipole (VED) at the origin in

$$\sigma(z) = \begin{cases} 0.1 \text{ S/m}, & z < 4 \text{ m}, \\ 1.0 \text{ S/m}, & z \geq 4 \text{ m}, \end{cases} \quad f = 2500 \text{ Hz},$$

with receivers on the z -axis, $z \in [0.5, 7.5]$ m, straddling the contact. The reference is the exact Sommerfeld-integral solution for E_z in a two-layer conductor (implemented in `analytics.py` and independently verified: it reduces to the closed-form homogeneous solution when $\sigma_1 = \sigma_2$ to 3×10^{-14} relative, and satisfies the normal-current continuity $\sigma_1 E_z^- = \sigma_2 E_z^+$ at the contact).

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Grid. Optimal geometric transverse grids with $k = 3$ ($M_x = M_y = 12$), a hybrid axial grid with fine equidistant spacing $\Delta z = 0.0625$ m near the source ($M_z = 160$, $N_R = 13,604$, $\approx 41k$ complex degrees of freedom), piecewise-constant σ with the contact resolved by the layered media builder.

Results. Figure 1 shows $|E_z|$ on the axis, the pointwise deviation from the analytic solution, and the contact effect (ratio of the two-layer to the homogeneous response).

- In the conductive half-space ($z \geq 4$ m), across the $10\times$ jump, the RMS relative error of $\text{Re } E_z$ is **1.0%**.
- The overall RMS error (5.6%) is dominated by the near-source nodes $z \lesssim 1.7$ m, where the transverse domain is only $0.3\delta_1$ wide ($\delta_1 \approx 32$ m in the resistive layer); the *same* error is present in the homogeneous control run (RMS 5.7%), so the material interface itself contributes essentially nothing — the FD scheme crosses the contact cleanly.
- The sharp physical structure at the contact is captured quantitatively: the two-layer/homogeneous ratio rises to 1.65 just above the contact (charge accumulation on the resistive side) and drops to ≈ 0.18 below it, matching the analytic curve point by point (Fig. 1, right panel).
- Cluster averaging matters: reading a single cluster (C000) gives RMS 29.9%; the four-cluster Lebedev average reduces this to 5.7% — the error-cancellation mechanism of DDH03 at work in an inhomogeneous medium.

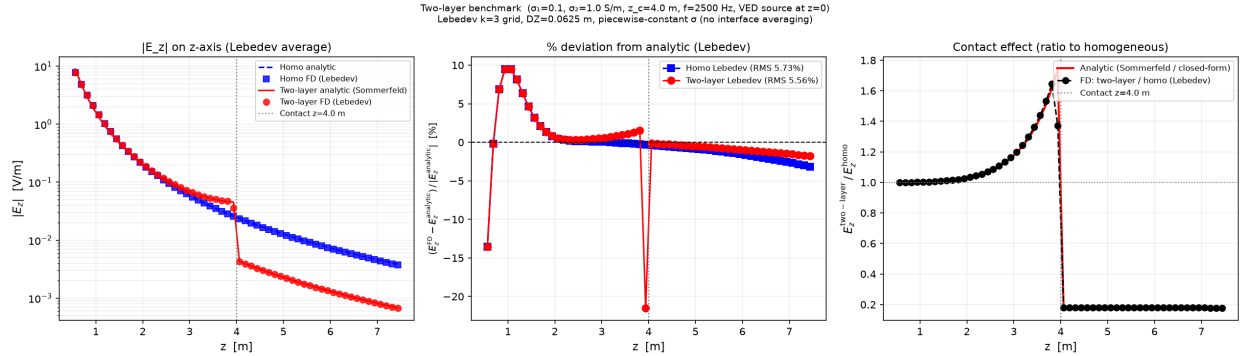


Figure 1: Two-half-space benchmark ($\sigma_1 = 0.1$, $\sigma_2 = 1.0$ S/m, contact at $z_c = 4$ m, $f = 2500$ Hz). Left: $|E_z|$ on the axis, FD (symbols) vs. analytic (lines), homogeneous control in blue, two-layer in red. Center: pointwise percent deviation from the analytic solutions. Right: contact effect — the ratio of two-layer to homogeneous E_z , FD (black) vs. Sommerfeld analytic (red). The isolated spike at $z = 4$ m in the center panel sits at the contact node itself, where E_z is discontinuous and a pointwise comparison is ill-posed.

3 Benchmark 2: DDH03 Figure 7 — borehole, invasion, dipping anisotropic layer

Model (DDH03 Fig. 6). The paper’s flagship 3-D inhomogeneous model, fully anisotropic and with no closed-form solution:

Borehole	$\sigma = 0.05$ S/m, radius 0.1 m
Invasion zone	$\sigma = 0.10$ S/m, radius 0.6 m
Dipping anisotropic layer	$\sigma_T = 0.1$, $\sigma_N = 0.01$ S/m, dip 60° , crossing the axis at $z = -0.5$ m
Isotropic layer	$\sigma = 0.5$ S/m
Source	x -directed magnetic dipole, 52.65 kHz, at the origin

The conductivity tensor of the dipping layer has its principal axes tilted 60° from vertical, so all off-diagonal couplings are active — the case Lebedev’s grid exists for. Cells cut by the borehole wall, invasion front, or dip plane are homogenized by sequential nodal homogenization with analytical interface normals.

Test. As in the paper, this is a *self-consistency convergence* test: the model is solved on optimal transverse grids with $k = 3, 4, 5, 6$ steps per half-axis ($M_x = M_y = 4k$; hybrid axial grid with $M_z = 98$; from $\approx 31k$ to $\approx 123k$ degrees of freedom), and the on-axis magnetic induction components $\text{Im } B_{xx}$ and $\text{Im } B_{xz}$ are compared across k . DDH03 report (p. 1535) that their curves for $k = 4, 5, 6$ are “practically identical” with $k = 3$ “almost as good.”

Results. Figure 2 shows the reproduction; the four curves overlay to plotting accuracy for both components, including the off-diagonal B_{xz} , which vanishes identically in a homogeneous medium and is generated here purely by the dipping inhomogeneity. Quantitatively, against the $k = 6$ reference:

vs. $k = 6$	$\text{Im } B_{xx}$		$\text{Im } B_{xz}$	
	max	mean	max	mean
$k = 3$ ($M_x = 12$)	7.8%	4.4%	5.4%	3.3%
$k = 4$ ($M_x = 16$)	3.6%	2.2%	4.1%	3.1%
$k = 5$ ($M_x = 20$)	1.5%	0.8%	1.5%	1.2%

The deviations shrink monotonically with k , consistent with the exponential convergence the optimal grids are designed to deliver, and the $k = 3$ solution — only three geometric steps to each side of the borehole, $M_x = 12$ — is already within a few percent everywhere, mirroring the paper’s central claim that the scheme handles complicated 3-D anisotropic models with remarkably few transverse nodes. Solve times on this hardware ranged from 4 s ($k = 3$) to 118 s ($k = 6$) with a sparse direct solver.

4 Remarks

Both benchmarks were run on the repository state including the July 2026 fixes (correct magnetic boundary rows in the curl operator, coordinate-based cluster interpolation, corrected geometry straddle detection — the latter directly affects which near-axis cells are homogenized in Benchmark 2). The two benchmarks are complementary: Benchmark 1 provides an *external* analytic reference but a simple geometry; Benchmark 2 has no analytic solution but exercises the full anisotropic, multi-material machinery and confirms internal convergence. A natural next step is the 2C-40 sonde comparison of DDH03 Figure 5, for which quasi-analytic reference curves exist in the literature (Anderson et al. 2001).

DDH03 Fig. 7 — Convergence test: borehole + invasion + 60° dipping anisotropic layer
52.65 kHz x-directed magnetic dipole | $M_x = M_y = 4k$, $M_z = 98$

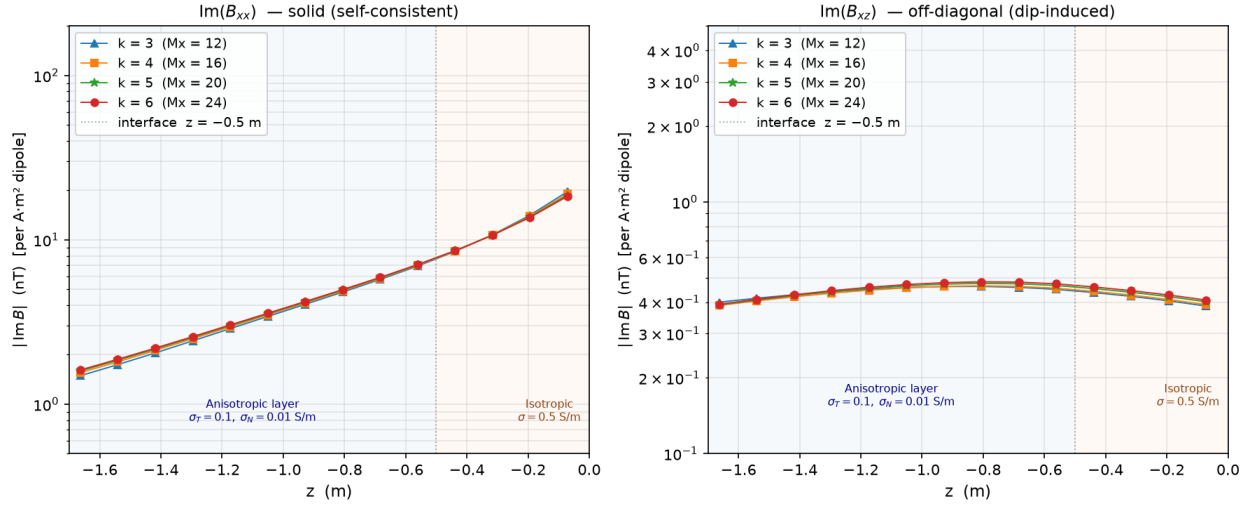


Figure 2: Reproduction of the DDH03 Figure 7 self-consistency test. $\text{Im } B_{xx}$ (left) and the dip-induced off-diagonal $\text{Im } B_{xz}$ (right) along the borehole axis for optimal grids with $k = 3, 4, 5, 6$; the interface of the dipping anisotropic layer crosses the axis at $z = -0.5$ m (dotted line).

References

- [1] S. Davdycheva, V. Druskin, T. Habashy, *An efficient finite-difference scheme for electromagnetic logging in 3D anisotropic inhomogeneous media*, Geophysics **68**(5):1525–1536, 2003.
- [2] S. Moskow, V. Druskin, T. Habashy, P. Lee, S. Davdycheva, *A finite difference scheme for elliptic equations with rough coefficients using a Cartesian grid nonconforming to interfaces*, SIAM J. Numer. Anal. **36**:442–464, 1999.